

Introduction to applied superconductivity

Experimental activity:
Measurement of an I-V curve in a coated conductor

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Part 1:

Quick refresh from lecture 2

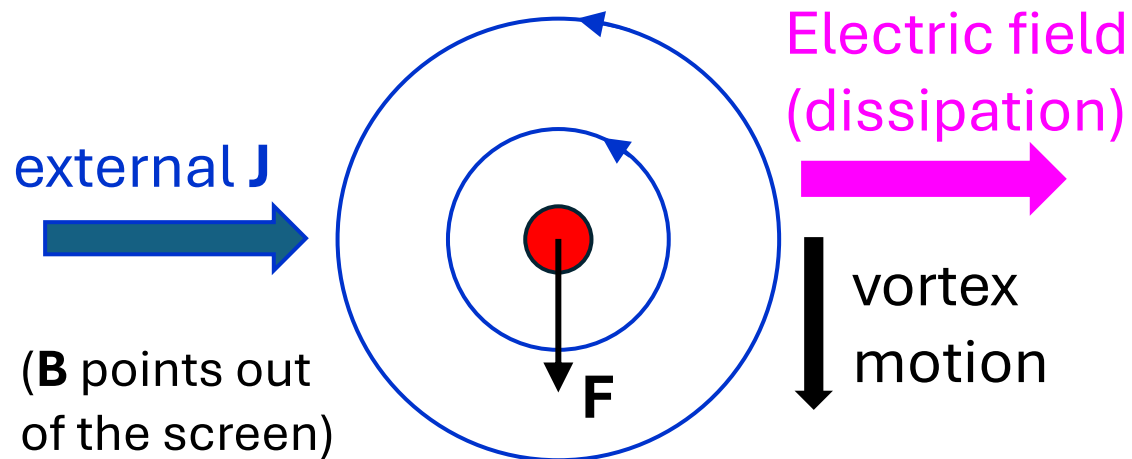
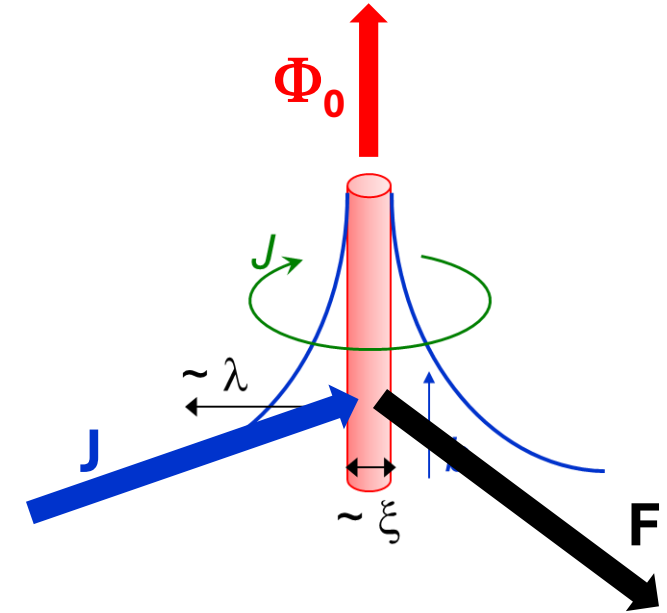
Forces that electric currents exert on vortices - flux flow

Force on the vortex

$$\mathbf{F} = \mathbf{J} \times \Phi_0$$

Vector of magnitude Φ_0 and direction of the vortex magnetic field

Electric current from any source other than the vortex (transport, Meissner screening, another vortex)



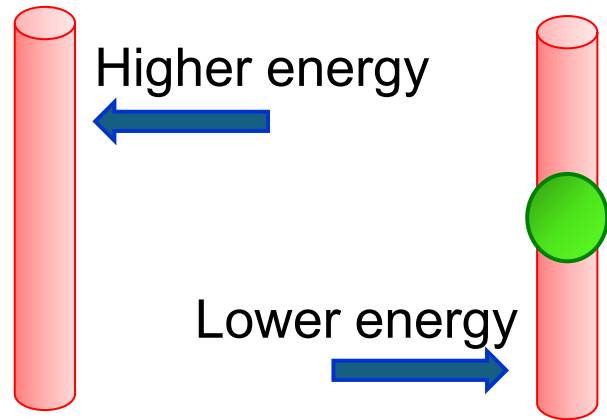
The problem:

An applied $\mathbf{J}$ exerts forces on the vortices $\Rightarrow$
 $\Rightarrow$ If vortices move, dissipation appears $\Rightarrow$
 $\Rightarrow$ Any benefit of a superconducting wire is lost!!

The solution: vortex pinning

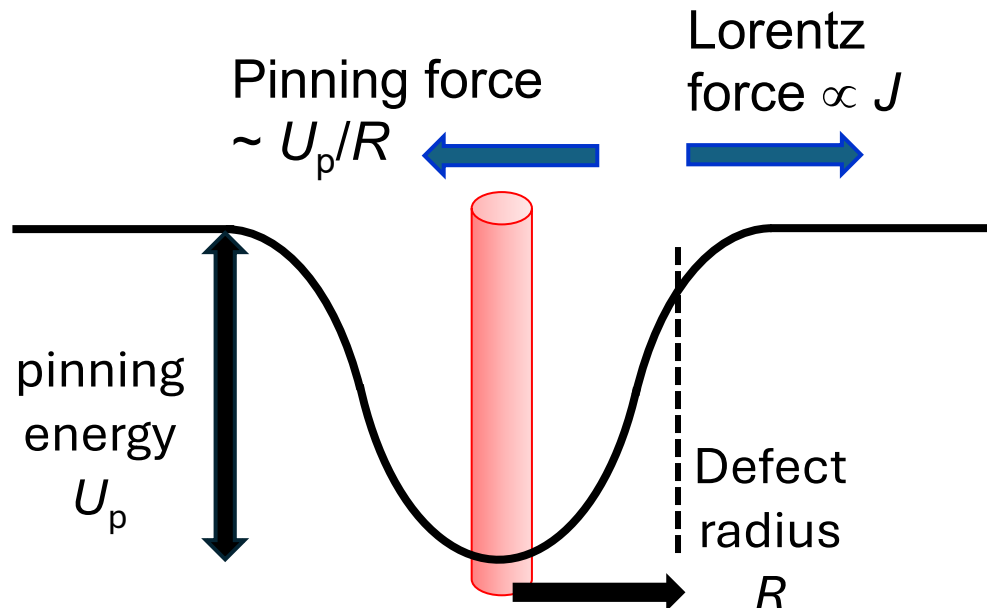
Vortex pinning

Vortex energy: ε_1 = magnetic + kinetic of the currents + condensation energy lost inside the core

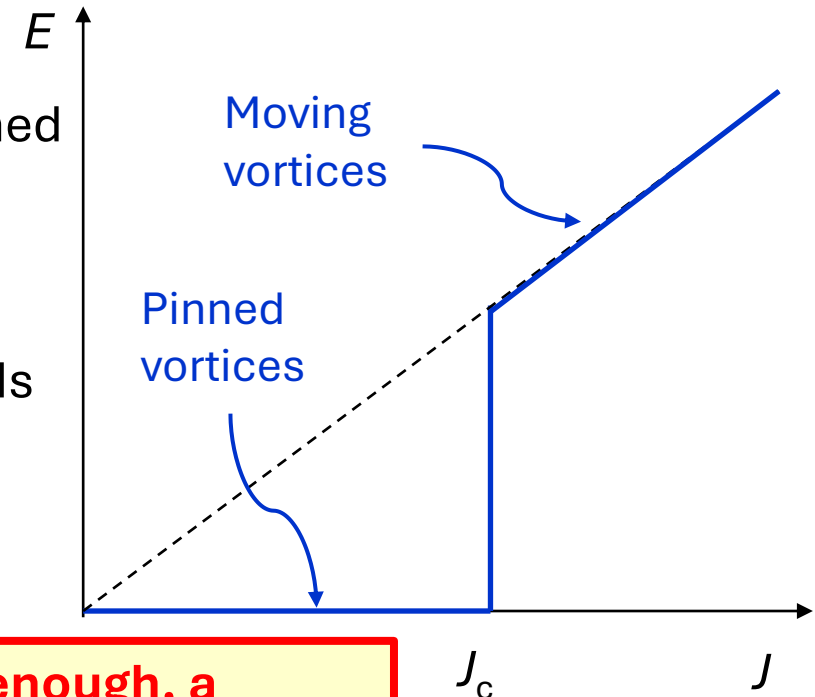


If a vortex sits on a portion of the material that is not superconducting (a defect), this energy cost disappears $\Rightarrow$ vortex costs less energy
The energy reduction is called **pinning energy**

Effect of an applied current:



Vortices remain pinned until J reaches the **critical current density** J_c , where Lorentz force exceeds maximum pinning force



If J_c is not high enough, a superconducting wire is useless

J_c depends on T, B, and field orientation

CC can operate
at very high
fields

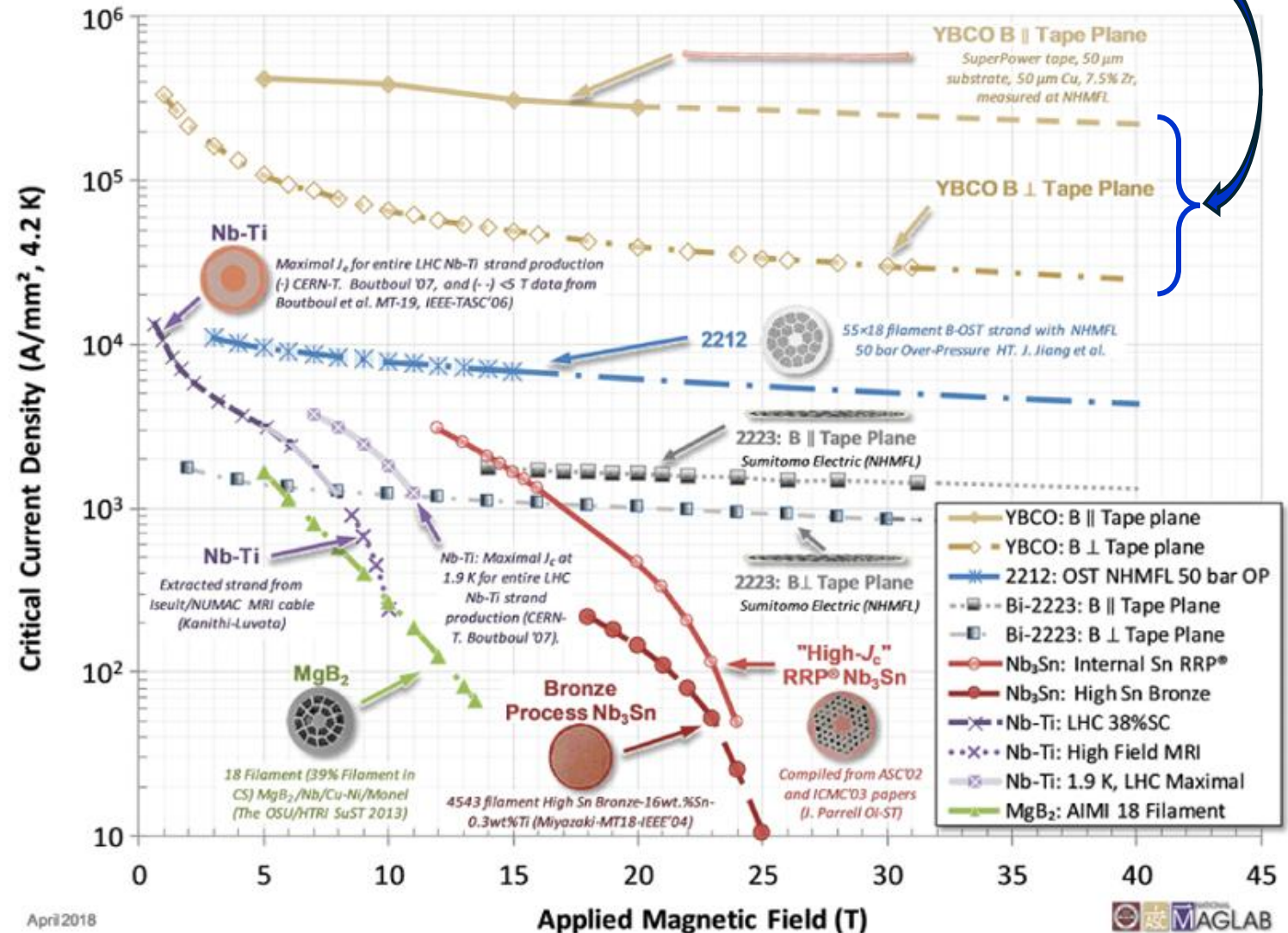
Usually, J_c decreases with T because

- U_p (related to ε_1) decreases with T
- The core radius ($\propto \xi$) increases with T
- Thermal fluctuations of vortices

Usually, J_c decreases with B because

- There may not be enough defects
- The best defects are already taken!
- Thermal fluctuations of vortices

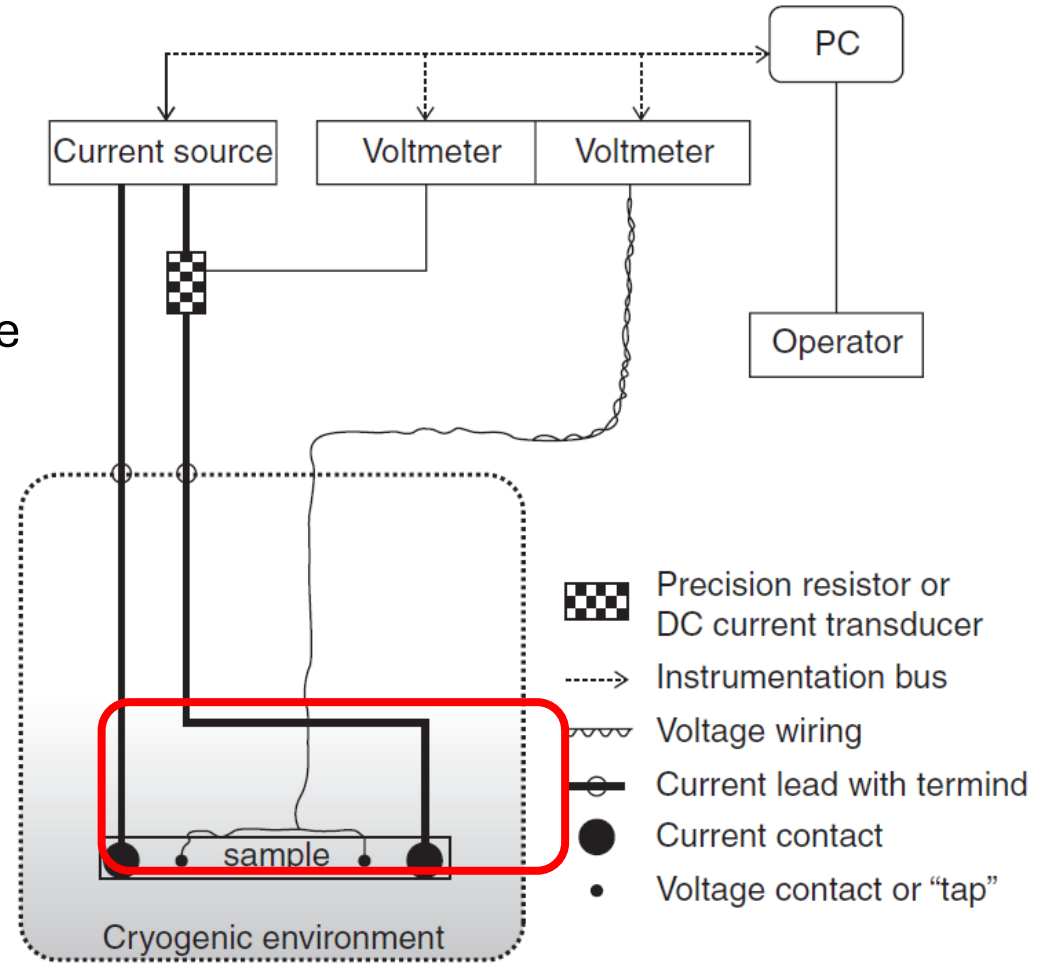
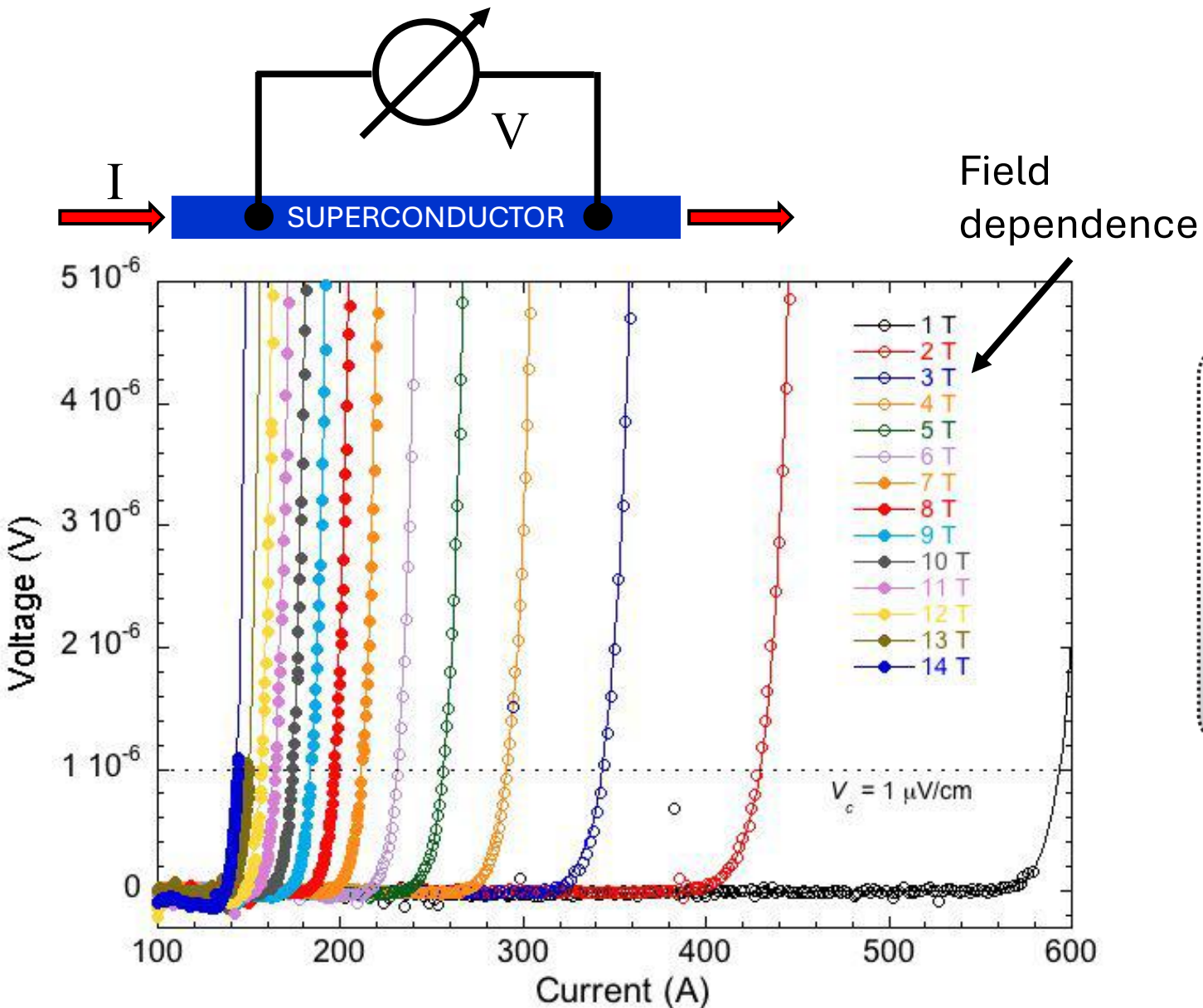
If the pinning is directional (linear or planar defects) and/or the material is anisotropic, J_c depends on the field orientation (more later)



April 2018

Methods to measure J_c

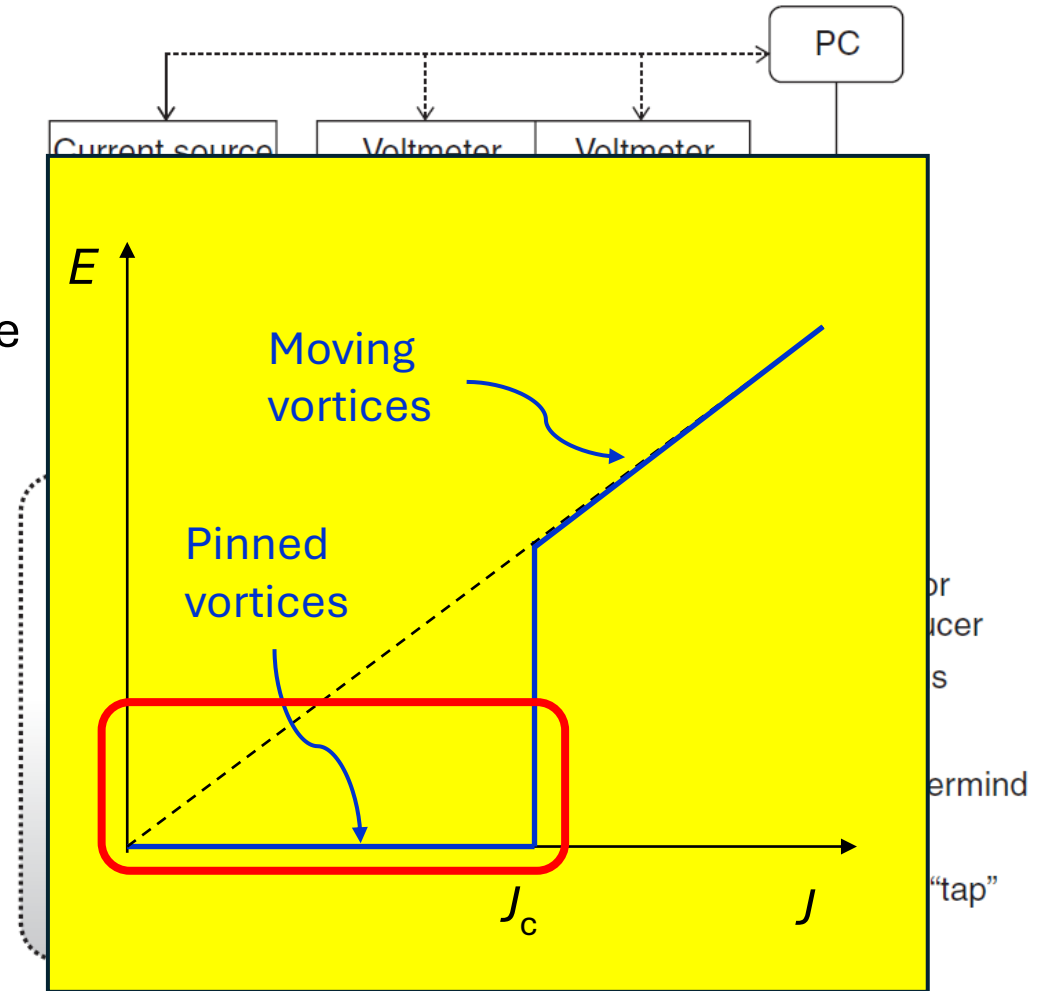
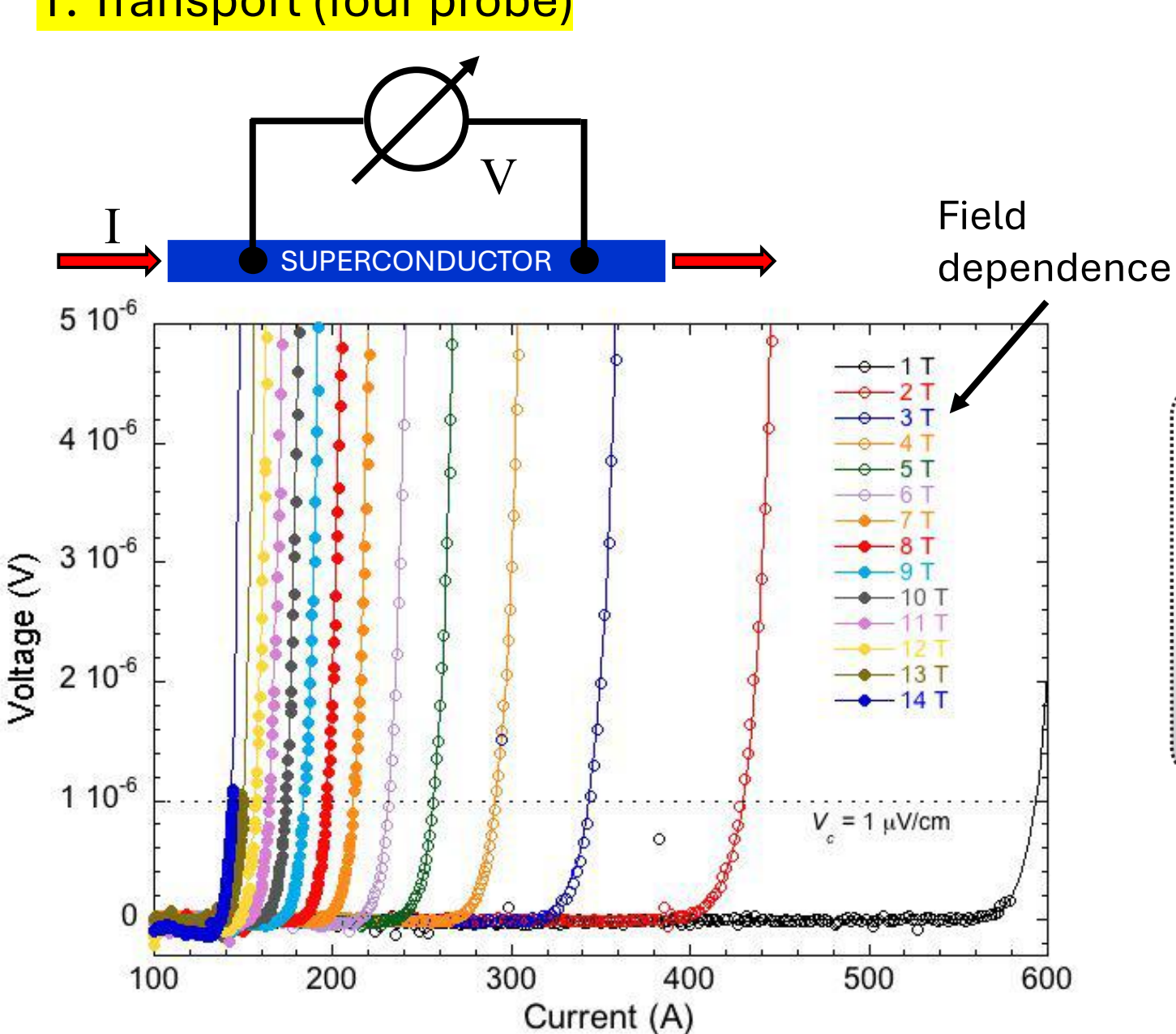
1. Transport (four probe)



These are the kind of measurements that I intend to do in the experimental part of this course

Methods to measure J_c

1. Transport (four probe)



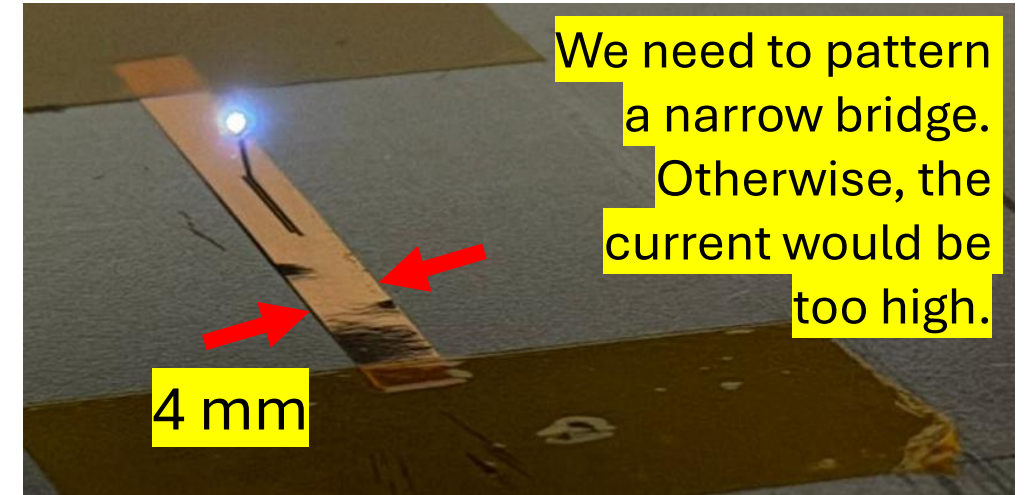
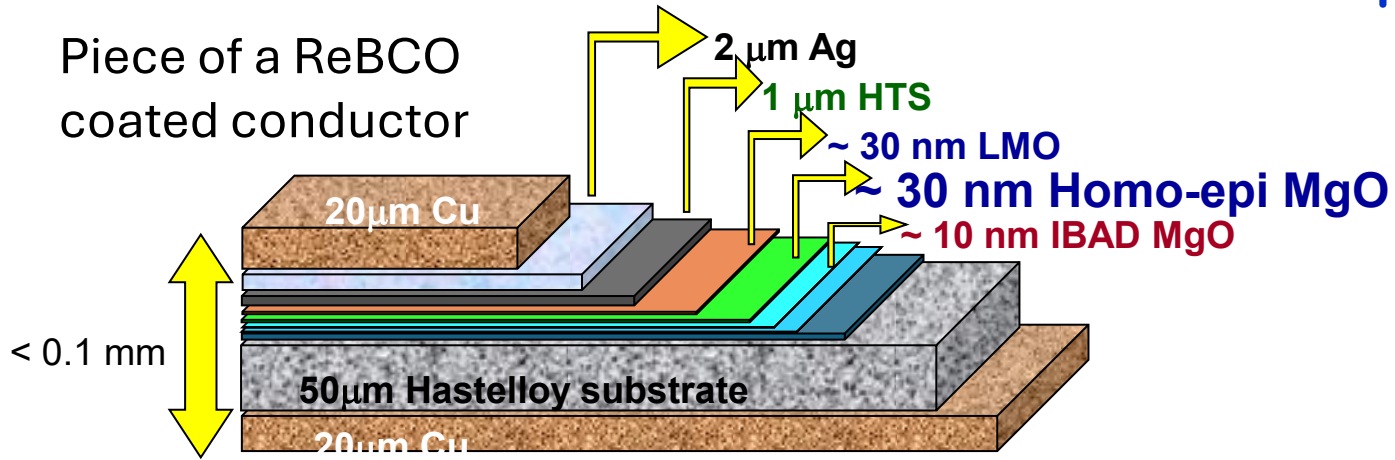
These are the kind of measurements that I intend to do in the experimental part of this course

Part 2:

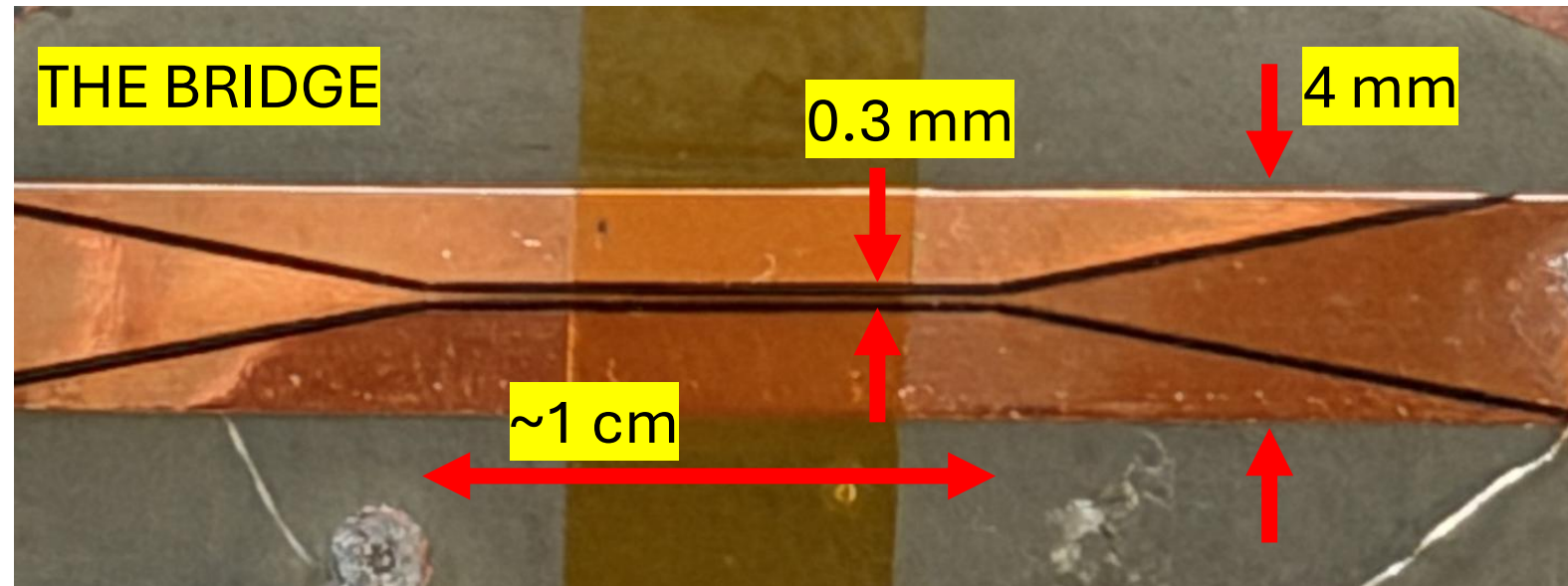
What we will do at the lab

The sample

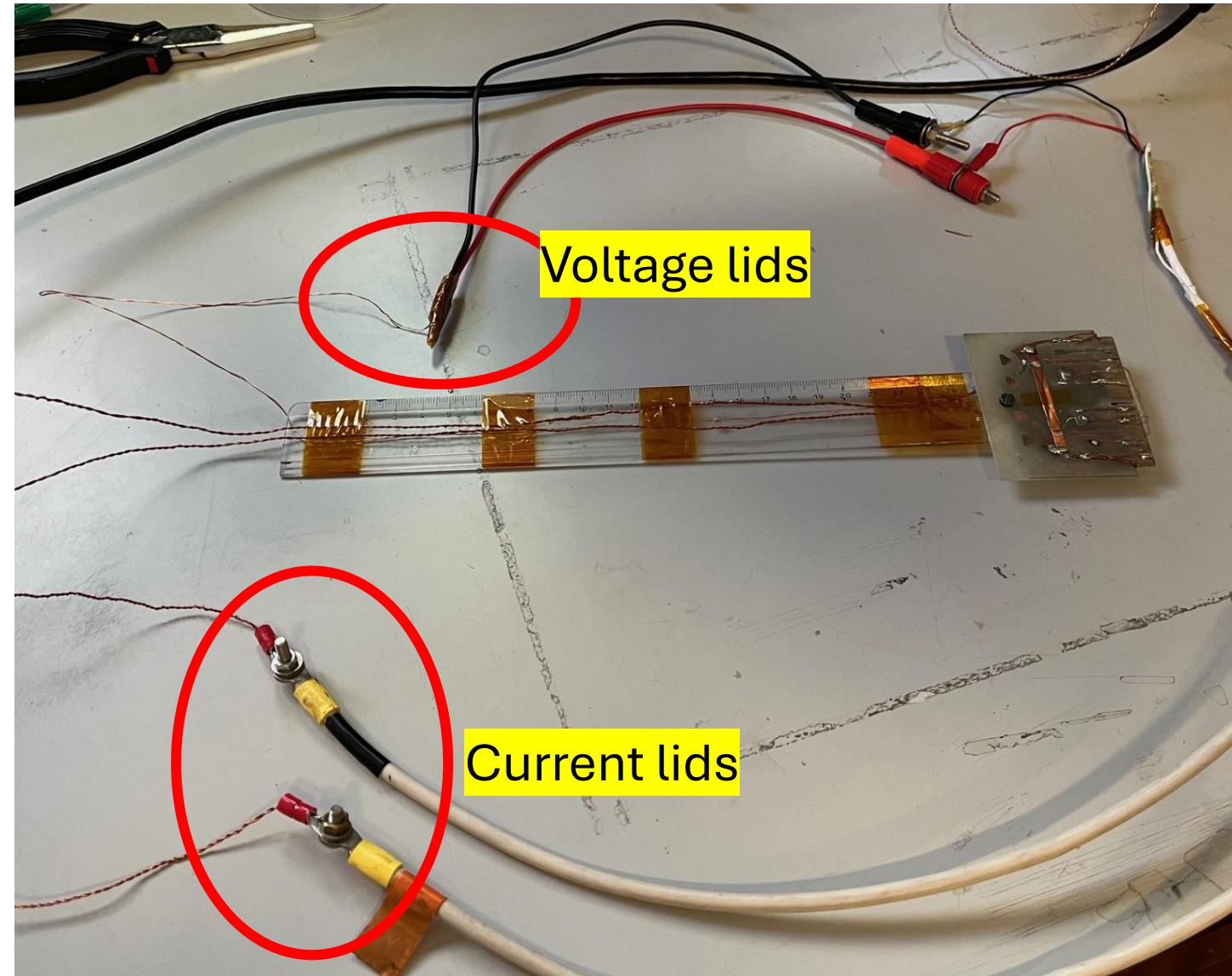
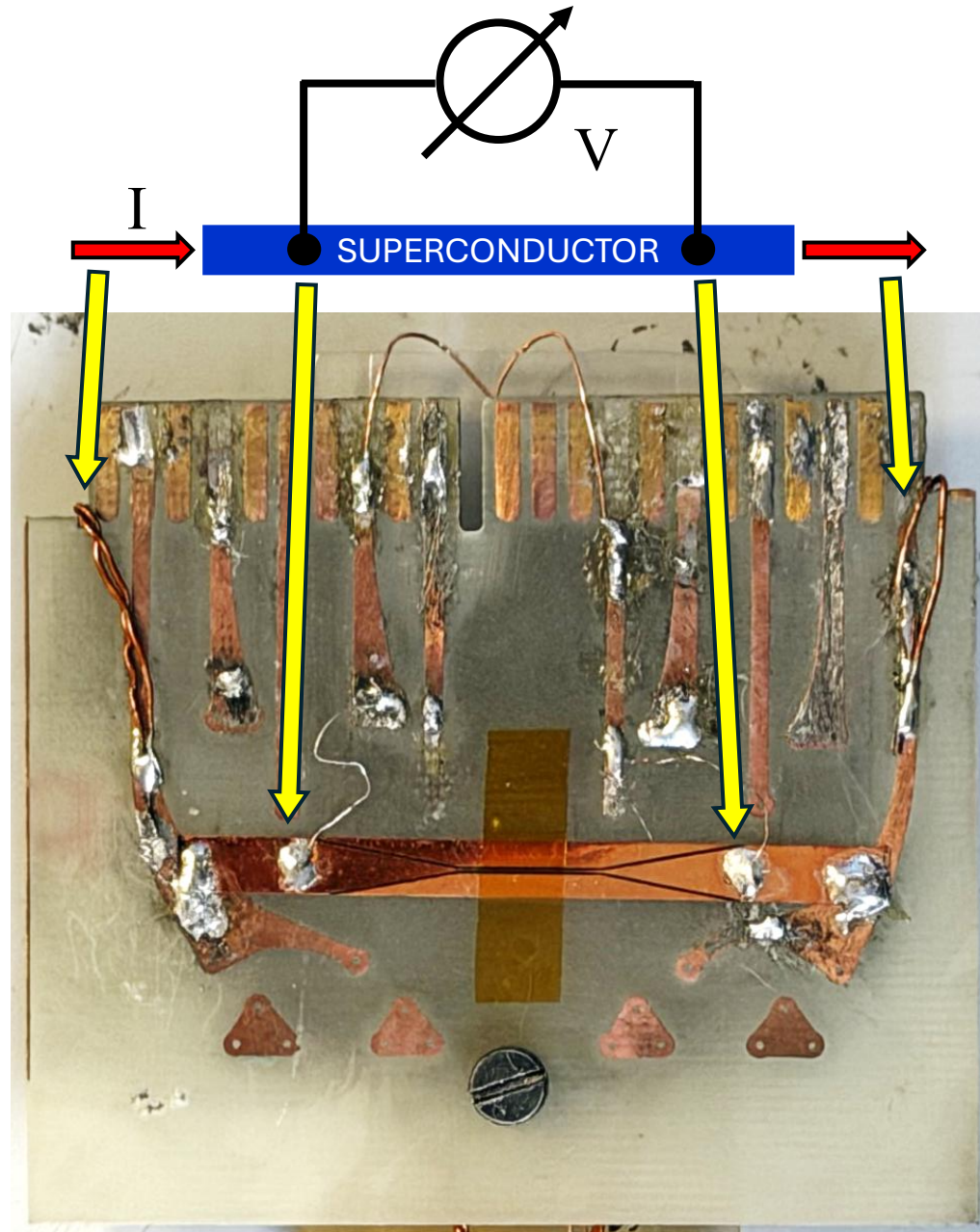
Piece of a ReBCO coated conductor



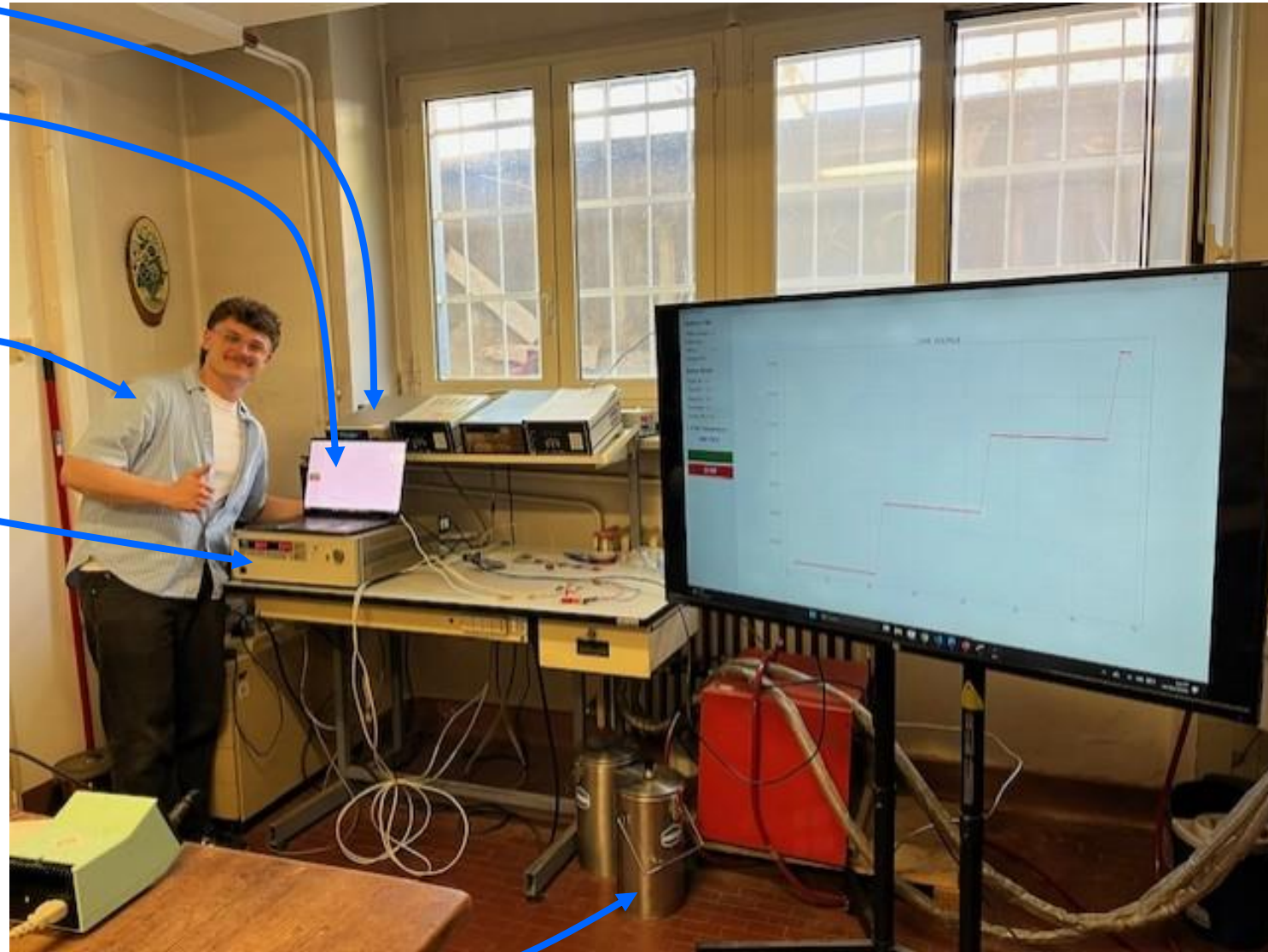
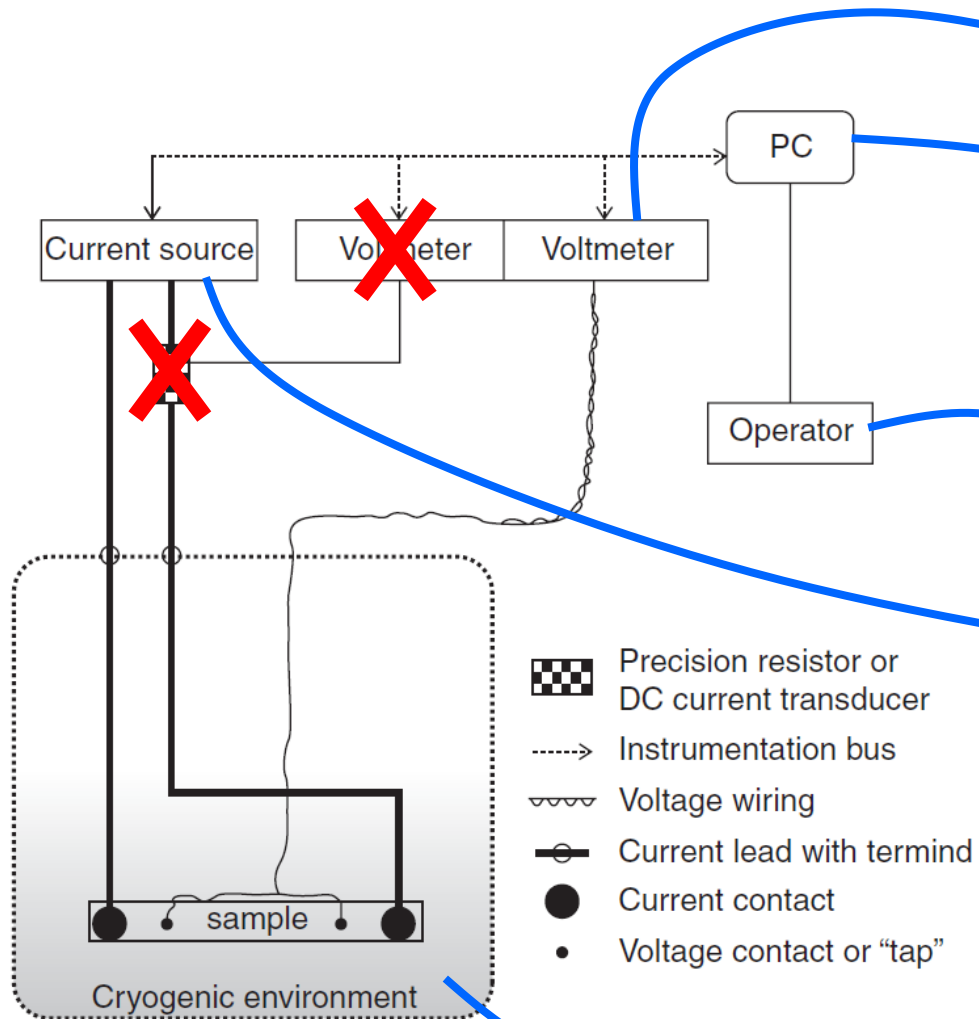
We use laser cutting



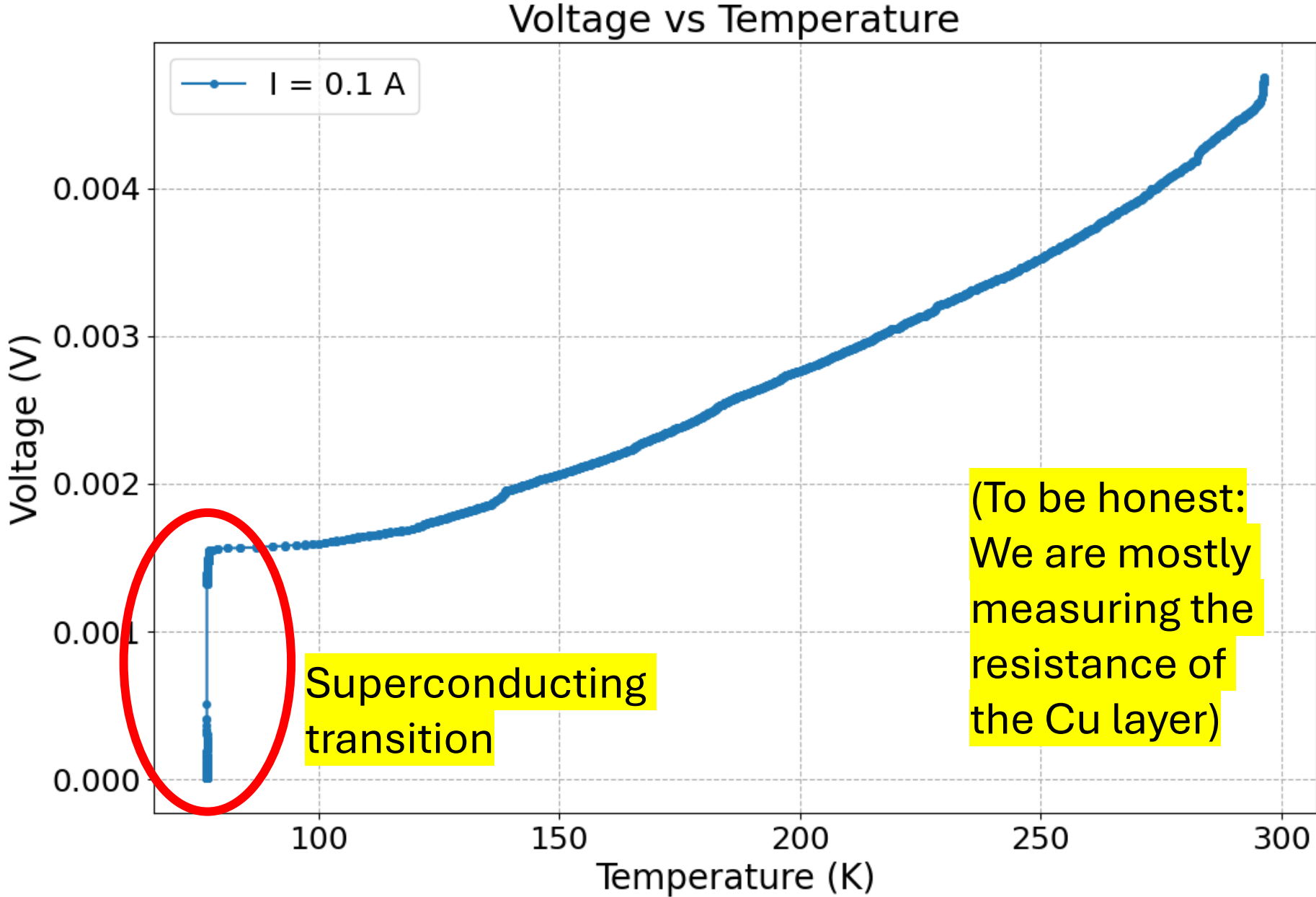
The wiring



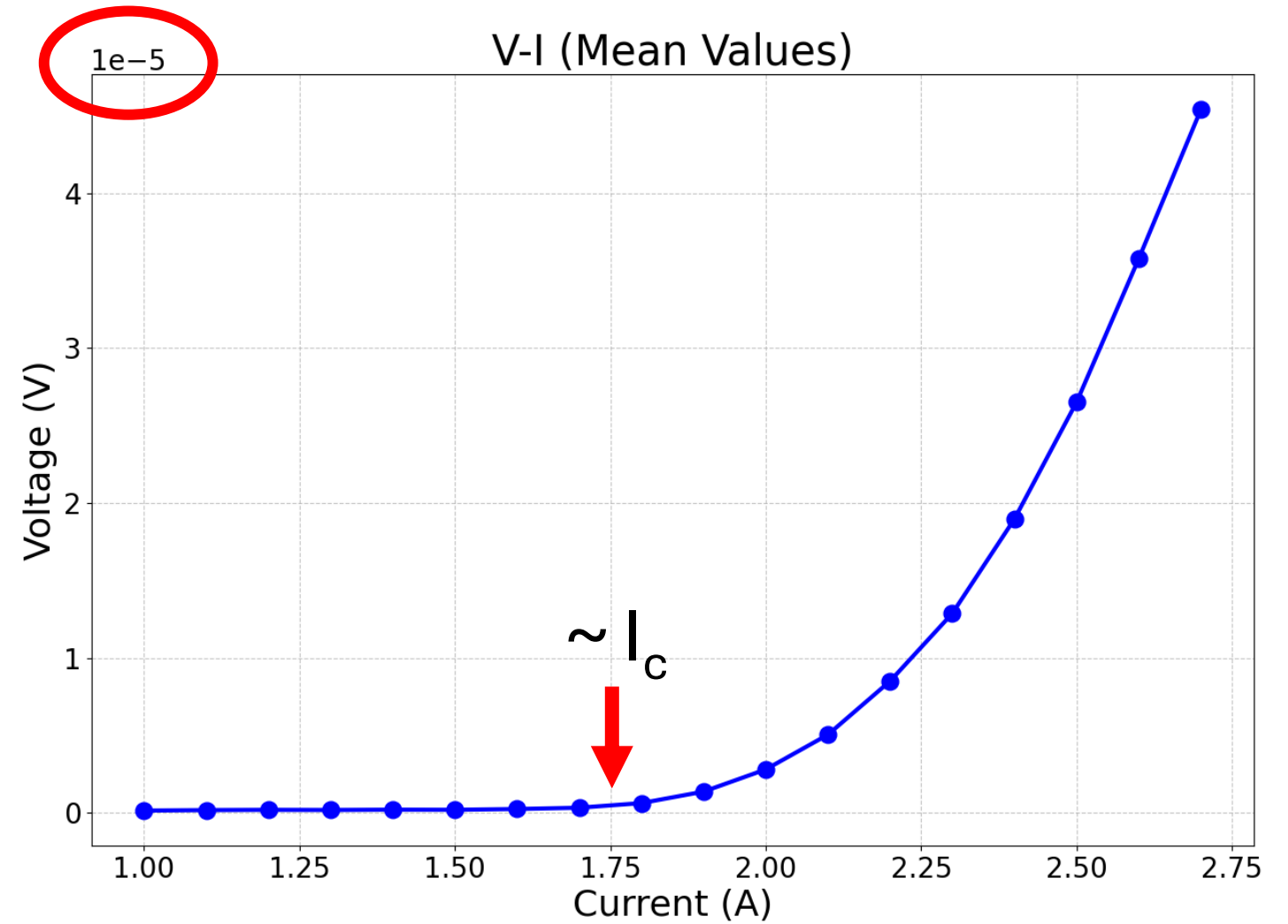
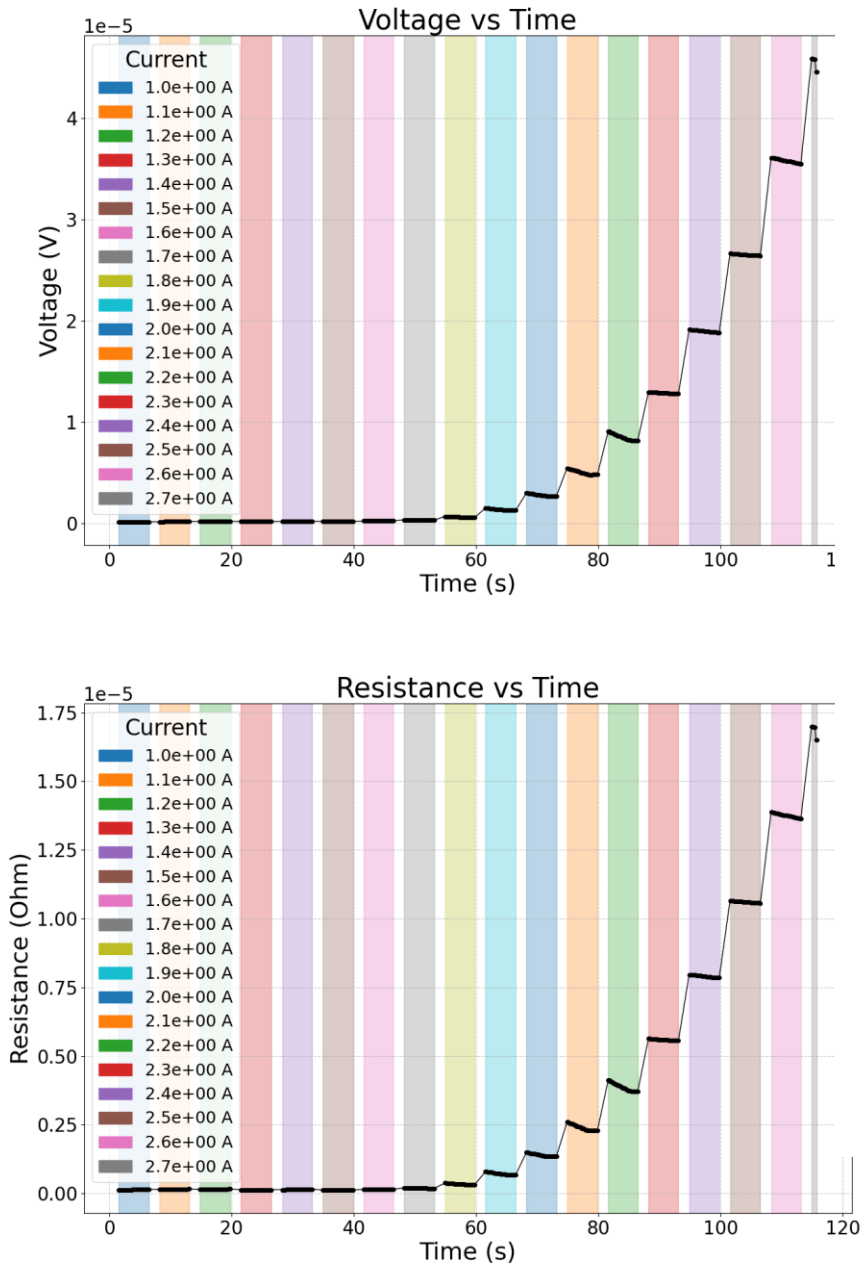
The measuring setup



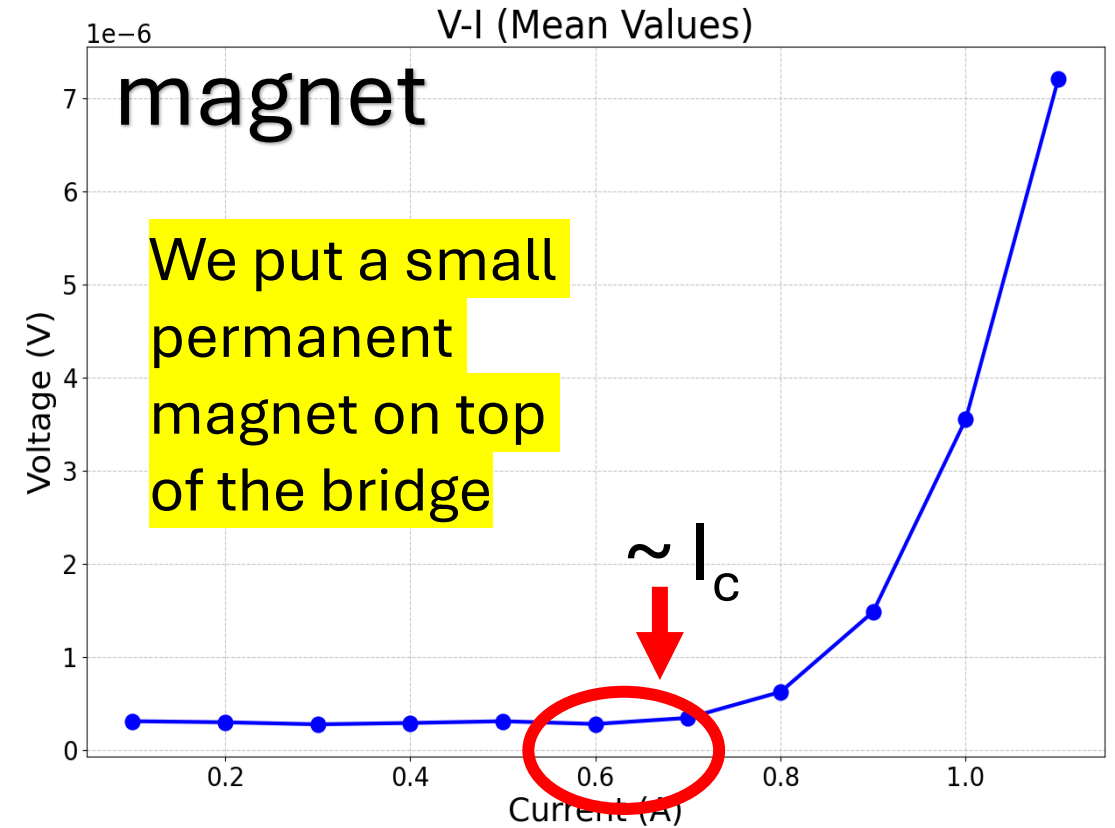
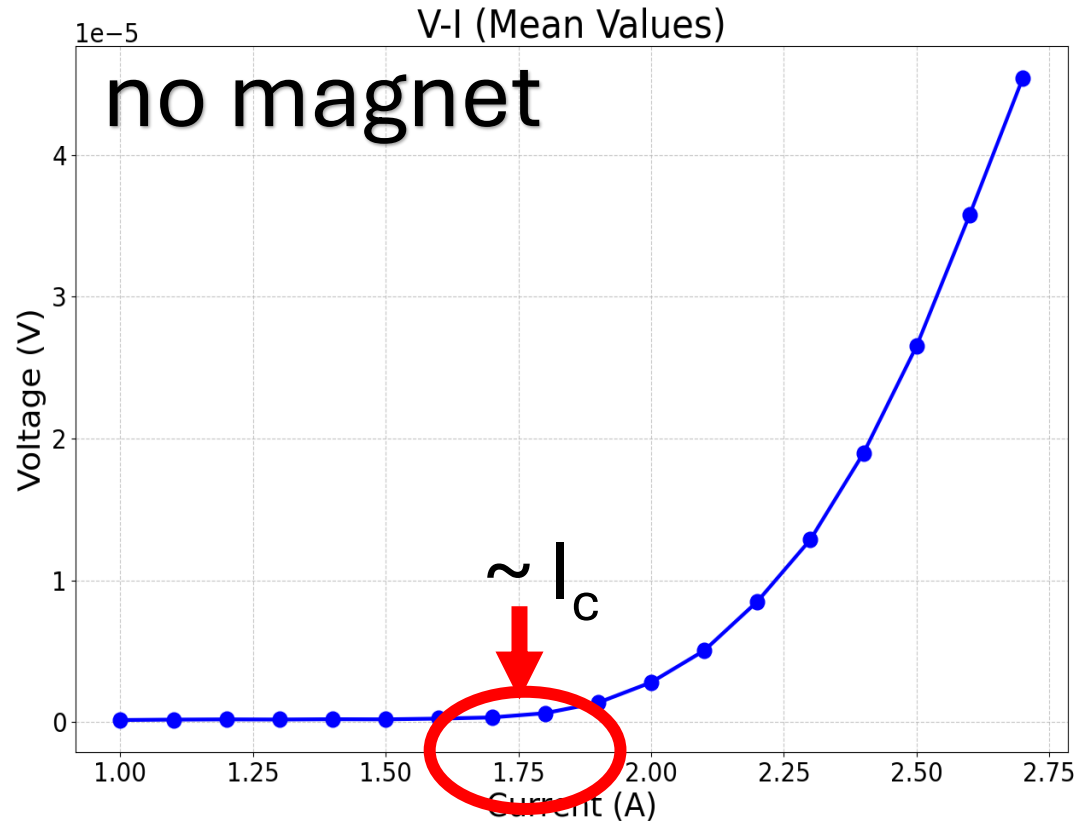
Tests results: temperature dependence of the resistance



Tests results: I-V curve



Tests results: field dependence of I_c



Grazie mille to the team!!

Special thanks to Simone Severo

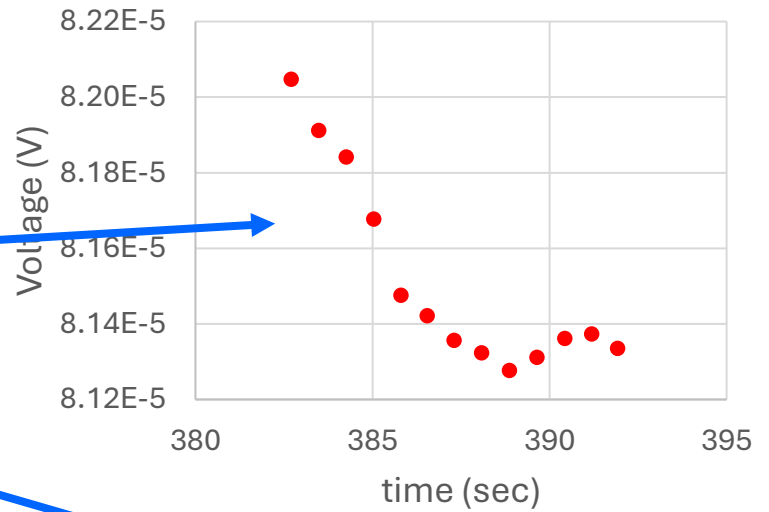
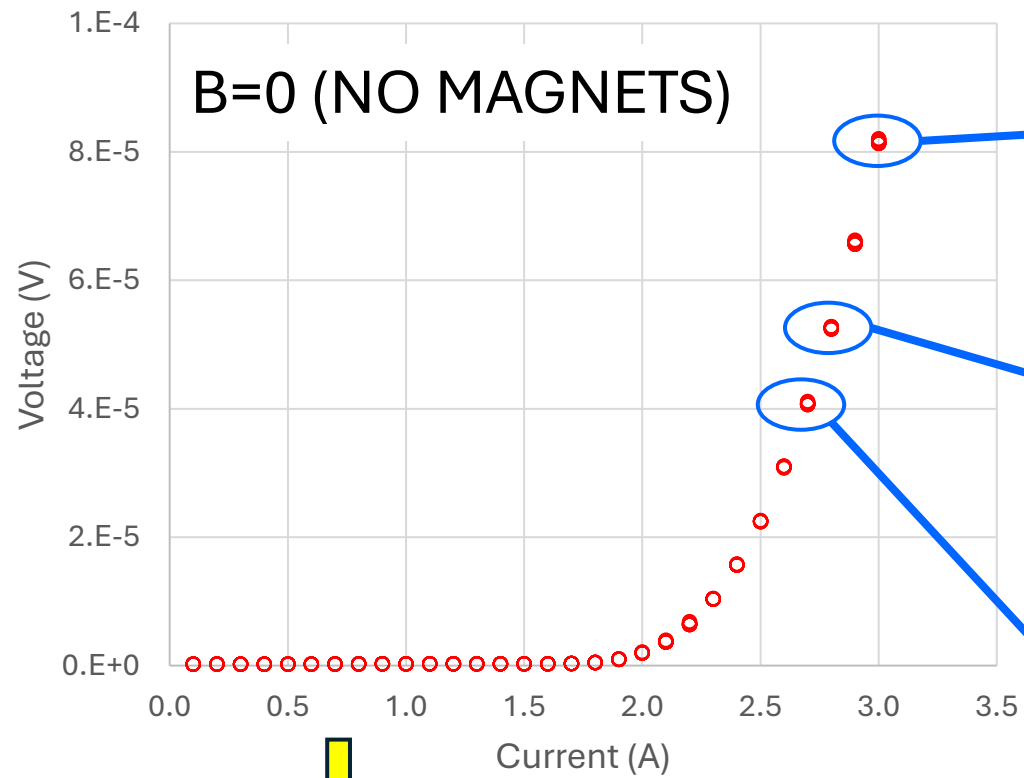
Thanks to Niccolò Gallino, Mattia De Stasio, Martina Casciello, Pasquale Petrillo, Laura Savoldi, Daniele Torsello, Francesco Laviano

Now, let's go to work...

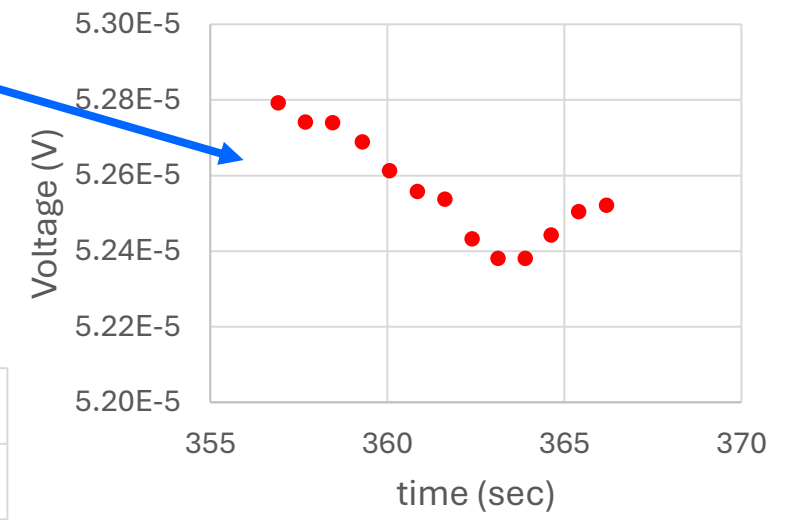
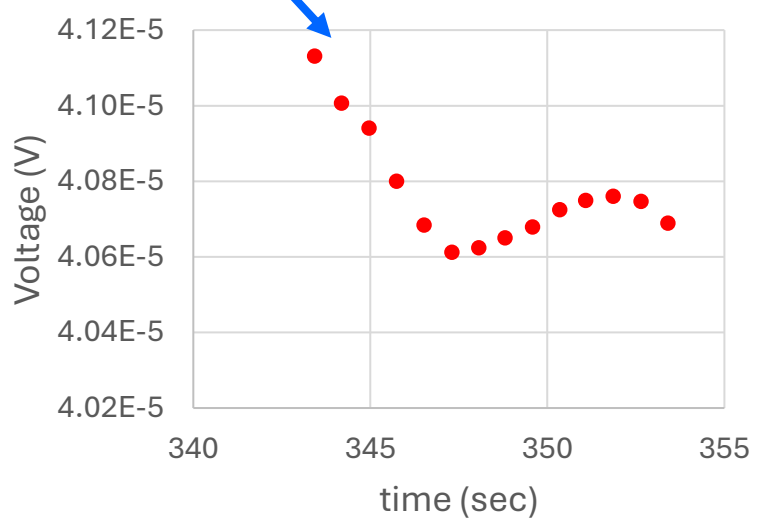
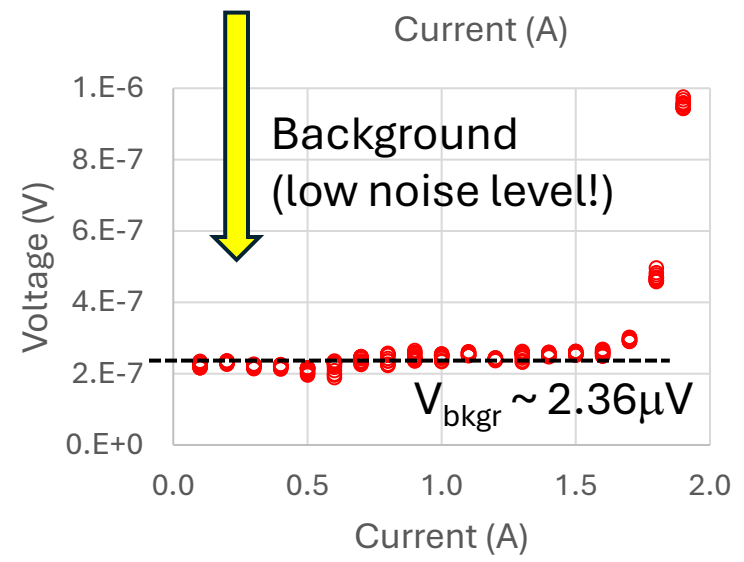
Part 3:

Analysis of the results

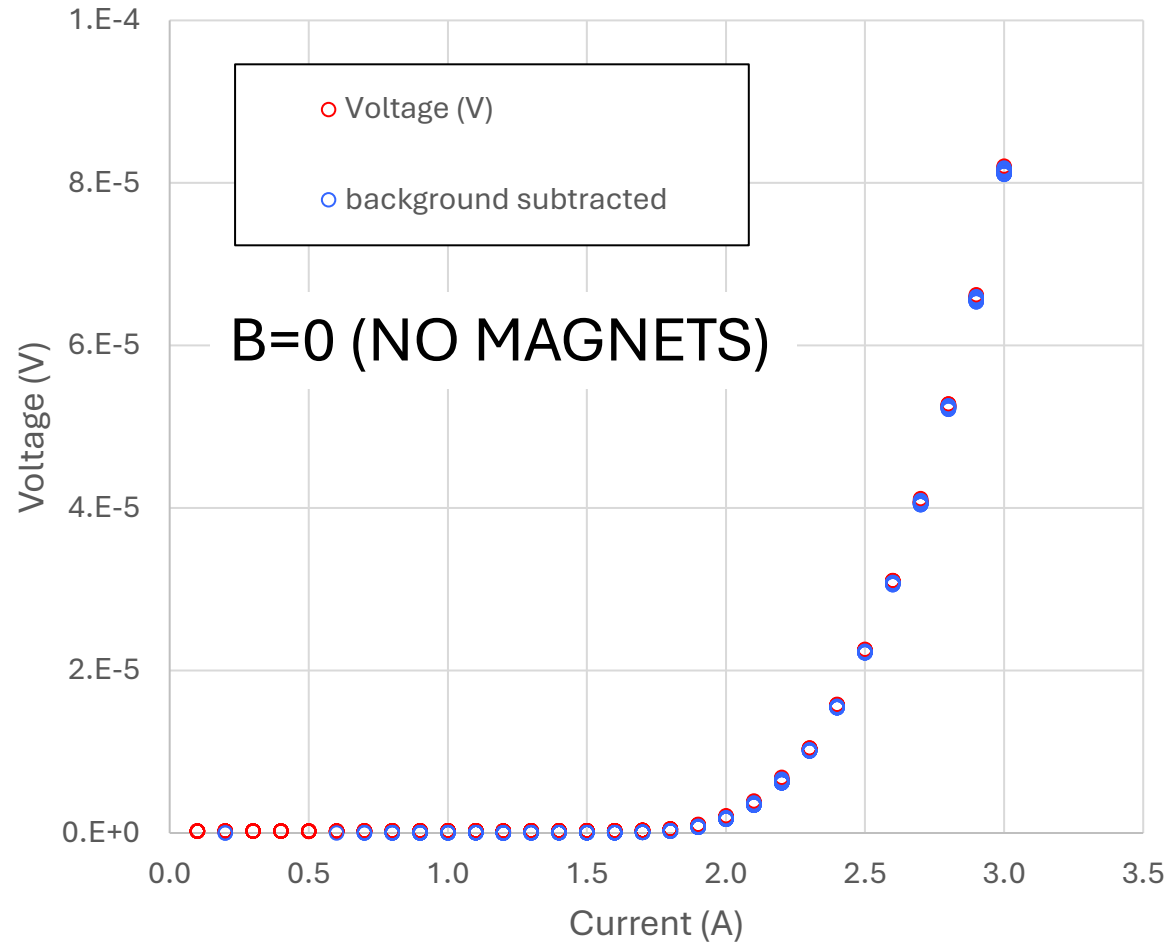
Analysis of an I-V curve



Initial voltage decay at each current is due to power supply transients
Electronics artifact



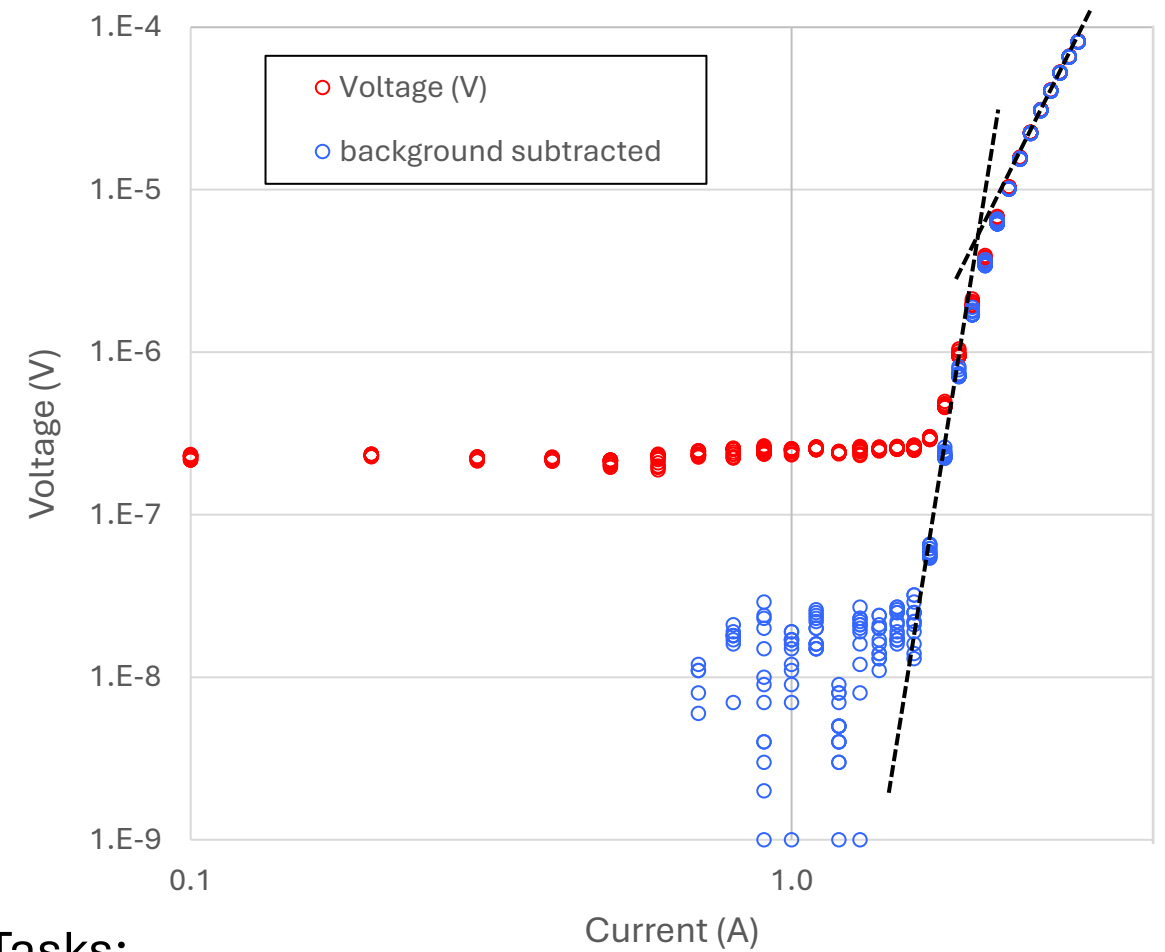
Analysis of an I-V curve



Expectation: power law $V \propto I^N$

$$V = V_c * (I/I_c)^N$$

Voltage criterion = 1 μ V/cm



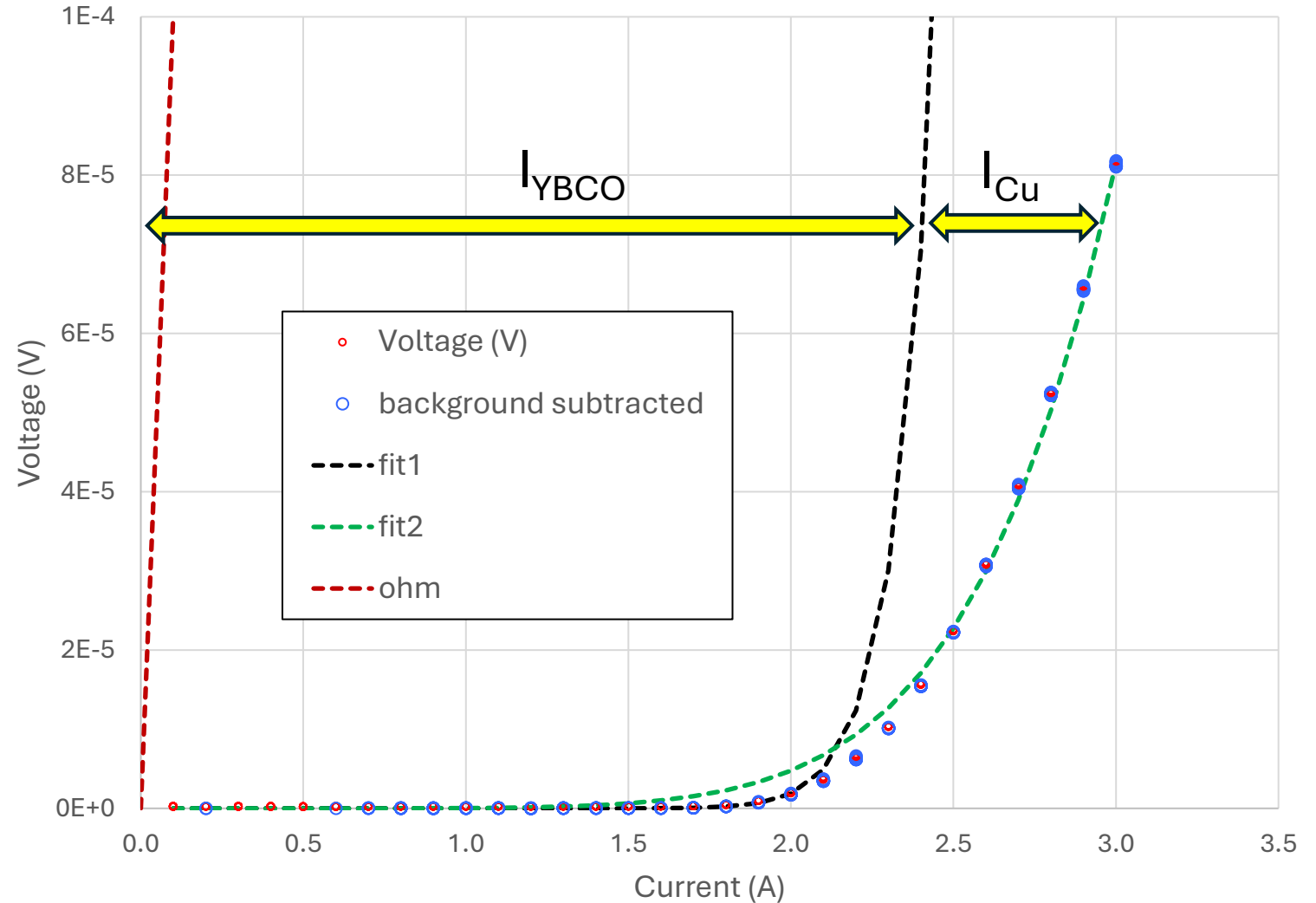
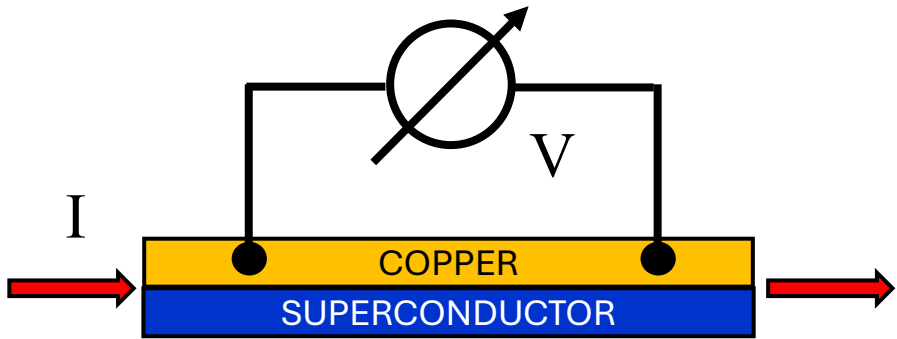
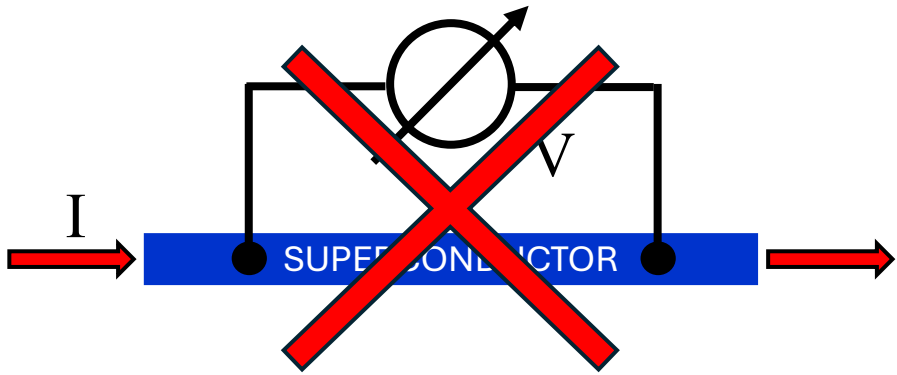
Tasks:

- Subtract background
- Fit initial power law, obtain I_c , N_1
- Fit second power law, obtain N_2

Additional task:

- Compare with I-V curves from other groups (minimum 5)
- Search for systematic degradation of I_c and N

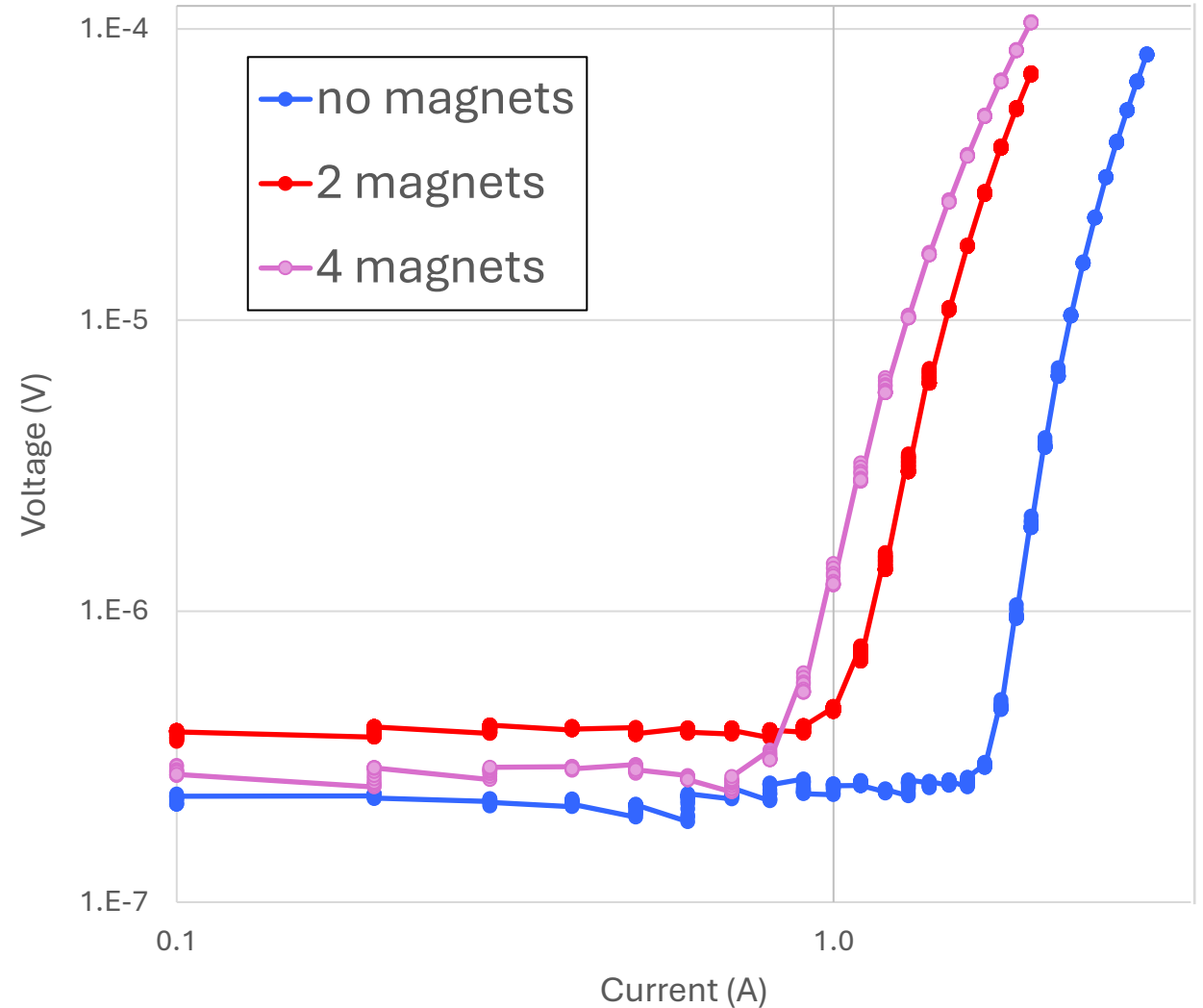
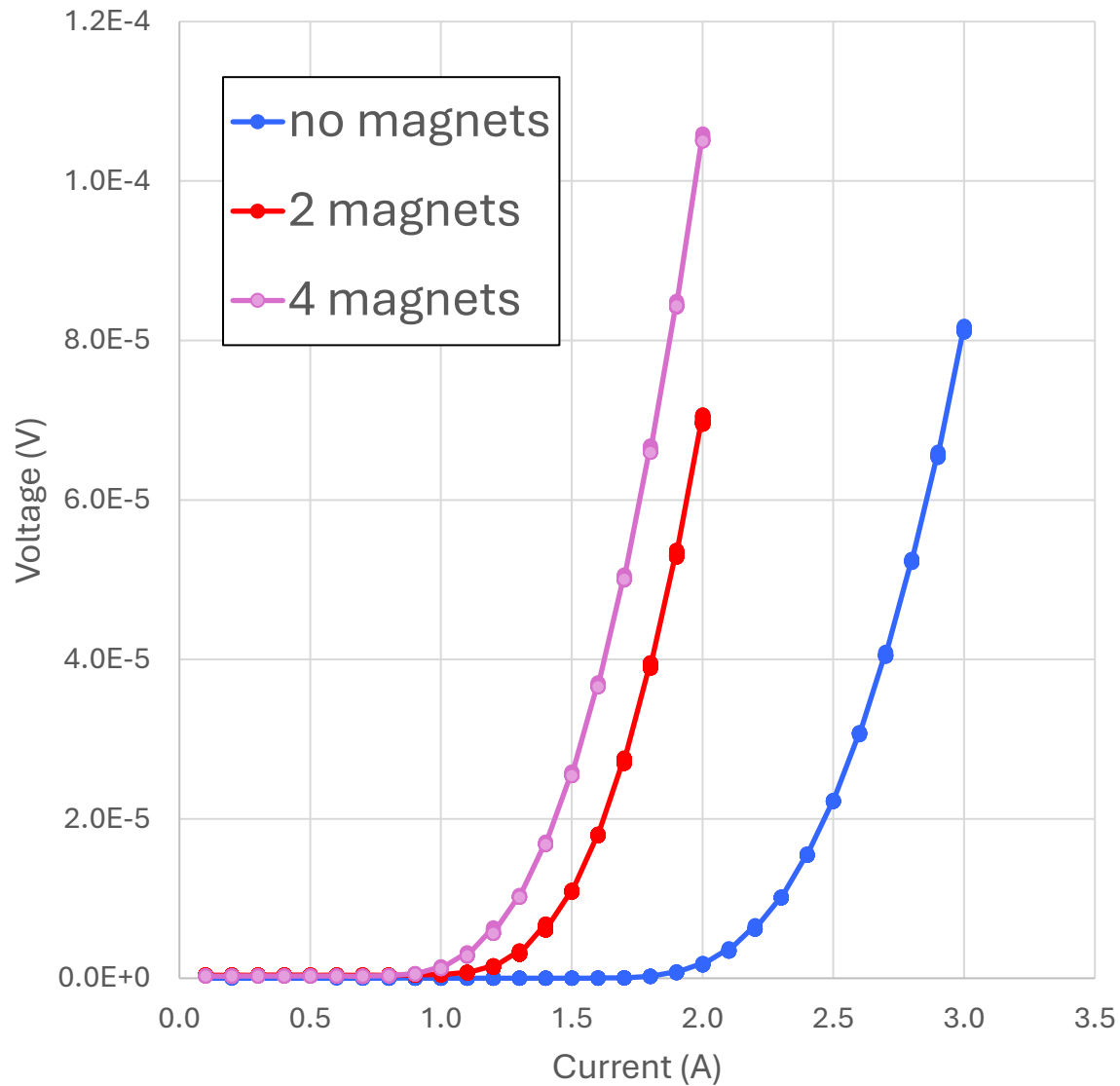
Effect of the copper layer?



When $V > 0$, part of the current starts to circulate through the copper

We approximately know R_{Cu} at 77K from the $R(T)$ data, but I did not obtain good quantitative agreement. Model was probably too simplified.

Effect of the magnetic field



Tasks:

- Repeat analysis for in-field curves
- Fit initial power law, obtain I_c , N_1